

# LEARNING SPATIAL PASS FEASIBILITY UNDER PARTIAL OBSERVABILITY IN SOCCER

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**ABSTRACT.** Pass outcome estimation in soccer is challenging due to partial observability in broadcast-derived data. This work studies whether meaningful pass probability surfaces can be learned from sparse, event-aligned freeze-frame observations using carefully designed spatial representations. Using the StatsBomb Open Dataset, we construct spatial state tensors composed of pitch geometry, player and ball encodings, and interaction-based control maps, and train a fully convolutional model under weak spatial supervision. Through controlled ablations and qualitative analysis, we show that pass completion conditioned on destination is robustly predicted, while destination selection remains inherently ambiguous. The results highlight the importance of state representation for interpretable soccer analytics under realistic data constraints.

## 1. INTRODUCTION

In sequential decision-making problems, the *state* describes the environment at a given moment and contains the information required to reason about future evolution. In chess, this state is fully observable: every piece and square is known, and the board configuration alone is sufficient to enumerate all legal future trajectories. Soccer, by contrast, admits no such compact representation. The true state of a soccer match includes continuous player and ball positions, motion, tactical structure, and latent intent, forming an effectively unbounded and partially observable system. To make analysis tractable, this complexity is compressed into discrete, processable representations. At the highest available resolution, modern spatiotemporal datasets approximate this ideal by capturing player and ball locations at high temporal frequencies, yielding near-complete coverage of the pitch and a sequence of annotated *actions* such as passes or shots.

This work is motivated by prior research conducted by the author and Ignacio Boero on offline Q-function estimation for soccer decision-making using the same underlying data source [1]. While that work demonstrated that long-horizon value estimates can be learned from high-dimensional spatiotemporal representations, it implicitly assumed access to a sufficiently complete and reliable state. In practice, limited observability and representation choices introduced errors that propagated into the learned value estimates, leading to instability and miscalibration that were difficult to diagnose. This revealed a more fundamental issue: stable long-horizon learning depends critically on what information the data can reliably support.

The present work revisits this issue from a more foundational perspective. We focus on freeze-frame data from the StatsBomb Open Dataset, which records partial snapshots of the field only at moments when an annotated action occurs. These snapshots omit large portions of play, exclude players outside the broadcast camera’s field of view, and provide no temporal continuity between events. As a result, the effective state available to the model is both spatially incomplete and temporally sparse (Figure 1). Rather than treating this limitation as prohibitive, this work asks a more focused question: how much information is required to recover meaningful spatial structure in passing behavior under partial observability? The primary contribution of this work is a qualitative study of pass probability estimation in this setting. Specifically, we (i) formalize the problem of learning dense pass completion probability surfaces from sparse, event-aligned broadcast data; (ii) propose and evaluate a set of interpretable spatial feature channels designed to compensate for missing players and incomplete field context; and (iii) study their effects through controlled ablation experiments with a fixed model architecture. Rather than introducing novel learning architectures, this work emphasizes how careful state construction enables meaningful spatial reasoning even when full spatiotemporal tracking data is unavailable, providing a foundation for future work on decision-making and value estimation in soccer. All code used for data processing, model training, and evaluation is publicly available at a GitHub repository.<sup>1</sup>

## 2. RELATED WORK

Early soccer analytics has largely focused on *scalar* event models with hand-crafted features, such as expected goals (xG), which map an action context to a single outcome probability. While effective for evaluating isolated events, these approaches do not produce *spatial* predictions over the pitch. More recent work has instead framed soccer as a spatial reasoning problem, enabling models to capture player positioning, space occupation, and interaction patterns across the field. A prominent line of work represents game states as image-like spatial tensors and applies convolutional neural networks to predict dense probability surfaces. SoccerMap [2], for example, encodes player and ball locations into multi-channel grids and predicts interpretable heatmaps indicating where actions such as passes are likely to succeed. Fernández et al. [3] extend this idea by decomposing value into action selection and state valuation, further emphasizing the benefits of spatially structured representations. Survey work by Pu et al. [4] documents a broader shift

<sup>1</sup><https://github.com/ethsen/Soccer-Research>

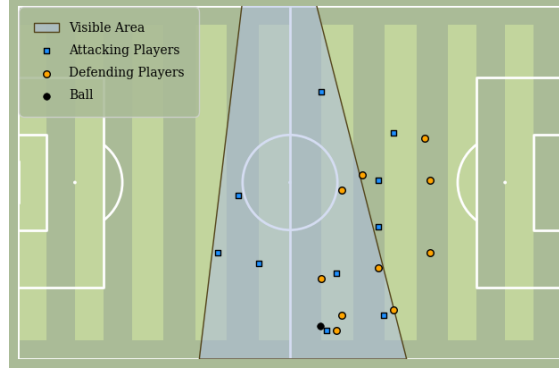


FIGURE 1. Broadcast visible area and player positions for a single frame, demonstrating the limited field of view available to broadcast-based soccer models. Attacking players (blue), defending players (orange), and the ball (black) are shown; only the shaded region corresponds to what is observable in the broadcast view.

in soccer analytics toward learning-based, spatially aware models that replace hand-crafted heuristics with data-driven representations of space and decision-making.

Several recent studies operate on the same StatsBomb 360 freeze-frame dataset used in this work, but pursue different objectives. Burriel and Buldú [5] propose a physics-inspired framework that evaluates pass quality by modeling interception risk under limited broadcast visibility. Robberechts et al. [6] introduce un-xPass, which assesses pass risk and creativity by comparing observed decisions to alternative options in similar contexts, focusing on evaluation rather than dense spatial prediction. Similarly, PassAI [7] predicts pass outcomes using engineered contextual features with an emphasis on interpretability, without operating on fully spatial, image-based representations.

In contrast to prior work that emphasizes predictive performance or architectural novelty, we focus on how *state representation* shapes learned pass probability surfaces under partial observability. By fixing the model architecture and varying the underlying spatial features, this study isolates which aspects of spatial context are most critical for robust and interpretable prediction when full spatiotemporal tracking data is unavailable.

### 3. PROBLEM SETTING AND STATE REPRESENTATION

We study pass probability estimation from sparse, broadcast-derived soccer data. Given a game state observed at the moment a pass is taken, the objective is to predict a dense spatial surface over the pitch indicating the probability that a pass directed to each location would be successfully completed. Each sample consists of a spatial state tensor  $s \in \mathbb{R}^{C \times H \times W}$  and a binary completion label associated with the observed pass destination. Learning is performed under weak spatial supervision, where only the grid cell corresponding to the realized destination is labeled.

**3.1. Dataset Processing and Pass Selection.** To reduce ambiguity arising from missing temporal context, we retain only passes labeled as completed when the ball’s recorded end location lies within 3 meters of a teammate. This restricts the task to short, directly receivable passes, where success can be inferred primarily from spatial geometry. All player and ball coordinates are mapped to a fixed top-down grid of size  $H \times W = 105 \times 68$ , and all channels are normalized to a common numeric range.

**3.2. Static Spatial Channels.** Static channels encode pitch geometry that is invariant across events and independent of player configuration. We include distance-to-boundary maps, distance to the centerline, and goal-oriented encodings such as distance and angle to the attacking goal. These channels provide global spatial priors that influence passing behavior and improve stability under partial observability by supplying structural information that would otherwise need to be inferred implicitly. Visualizations of these channels are provided in the appendix.

**3.3. Dynamic Spatial Channels.** Dynamic channels encode the observable game state at the time of the pass, including ball-relative features (distance and direction to the ball) and binary occupancy maps for attacking and defending players. We additionally include coarse motion summaries derived from event-to-event displacements. Because freeze-frame data does not support reliable identity tracking, per-player velocities are unavailable; instead, we approximate global attacking and defending team velocity using centroid displacement between consecutive events, and approximate ball

velocity using ball displacement over time. These motion features provide weak contextual cues but are not essential for accurate pass-probability estimation. Example visualizations are included in the appendix.

**3.4. Control Maps for Spatial Interaction.** To encode local spatial interaction under partial observability, we introduce two complementary control maps derived directly from observed player locations: a Gaussian control map and a nearest defender distance map. Both summarize how attacker and defender presence shapes pass feasibility without attempting to reconstruct unobserved players or temporal dynamics.

Let  $\mathcal{A}$  and  $\mathcal{D}$  denote the sets of observed attacker and defender grid locations. The Gaussian control map represents spatial influence by summing isotropic Gaussian kernels centered at each player location,

$$A_p(g) = \sum_{a \in \mathcal{A}} \exp\left(-\frac{\|g - a\|_2^2}{2\sigma^2}\right), \quad D_p(g) = \sum_{d \in \mathcal{D}} \exp\left(-\frac{\|g - d\|_2^2}{2\sigma^2}\right). \quad (1)$$

These fields provide smooth estimates of attacking and defending presence, but their absolute scale depends on the number and spatial concentration of observed players. We therefore use them only to construct a relative control estimate rather than interpreting them directly as probabilities. Specifically, we first form a smoothed dominance ratio,

$$\tilde{C}(g) = \frac{A_p(g) + \alpha}{A_p(g) + D_p(g) + 2\alpha + \varepsilon}, \quad (2)$$

where  $\alpha, \varepsilon > 0$  ensure numerical stability and prevent extreme values when few or no players are observed.

Because broadcast-derived freeze-frames often contain large regions of the pitch with little or no player information, we further modulate this dominance estimate using a relative evidence weight. We define

$$e(g) = \frac{A_p(g) + D_p(g)}{\max_{g'} (A_p(g') + D_p(g')) + \varepsilon}, \quad (3)$$

which measures local player density relative to the most densely observed region in the current frame. This normalization is performed independently for each event, making  $e(g)$  a purely relative measure of observational support rather than an absolute density estimate.

The final Gaussian control map is obtained by interpolating between the dominance estimate and a neutral prior,

$$C(g) = e(g) \tilde{C}(g) + (1 - e(g)) \cdot 0.5. \quad (4)$$

In regions with high observed player density,  $C(g)$  reflects the local balance of attacking and defending influence; in sparsely observed regions, it smoothly reverts to an uninformative value of 0.5, preventing spurious confidence in areas where the state is poorly observed. As a simpler and more interpretable alternative, the nearest defender distance map encodes defensive pressure as the Euclidean distance to the closest observed defender,

$$d_{\min}(g) = \min_{d \in \mathcal{D}} \|g - d\|_2, \quad (5)$$

converted to meters. This representation is robust to missing defenders and captures immediate local pressure without imposing assumptions about aggregate interaction structure. Together, these control maps provide complementary views of spatial interaction under partial observability: the Gaussian map captures smooth, relative dominance modulated by observational evidence, while the nearest defender map captures sharp local defensive constraints. Visualizations of both control maps are provided in the appendix.

## 4. MODEL AND TRAINING OBJECTIVE

**4.1. Learning Objective.** Given a spatial state representation  $s \in \mathbb{R}^{C \times H \times W}$  observed at the moment a pass is taken, the model predicts two dense spatial quantities: (i) a distribution over pass destinations across the pitch, and (ii) the probability that a pass to a given location would be successfully completed. The model outputs a destination logit map  $Z^{\text{dest}} \in \mathbb{R}^{H \times W}$  and a success logit map  $Z^{\text{succ}} \in \mathbb{R}^{H \times W}$ , trained using weak supervision from the observed pass destination and completion label.

**4.2. Destination Distribution Loss.** Rather than supervising destination prediction with a single positive grid cell, we define a smooth target distribution centered at the observed destination  $(x^*, y^*)$  using a Gaussian kernel,

$$P^*(g) \propto \exp\left(-\frac{\|g - (x^*, y^*)\|_2^2}{2\sigma^2}\right), \quad (6)$$

normalized over the  $H \times W$  grid. The destination loss is defined as the Kullback–Leibler divergence between the target distribution and the model’s predicted distribution,

$$\mathcal{L}_{\text{dest}} = \text{KL}(P^* \parallel \text{softmax}(Z^{\text{dest}})), \quad (7)$$

encouraging smooth and spatially coherent destination surfaces.

**4.3. Pass Success Loss.** Pass completion is supervised only at the observed destination cell. Let  $y \in \{0, 1\}$  denote the binary completion label. The success logit at  $(x^*, y^*)$  is trained using binary cross-entropy,

$$\mathcal{L}_{\text{succ}} = \text{BCE}(\sigma(Z^{\text{succ}}(x^*, y^*)), y), \quad (8)$$

modeling the probability of successful reception conditioned on a specific spatial target.

**4.4. Combined Objective.** The final training objective is a weighted sum of the two losses,

$$\mathcal{L} = \lambda_{\text{dest}} \mathcal{L}_{\text{dest}} + \lambda_{\text{succ}} \mathcal{L}_{\text{succ}}, \quad (9)$$

with  $\lambda_{\text{succ}}$  weighted more heavily to prioritize accurate estimation of pass completion probabilities, while  $\mathcal{L}_{\text{dest}}$  acts as a regularizer on the spatial structure of the destination distribution.

**4.5. Model Architecture.** All experiments use the same fully convolutional two-head architecture, consisting of a shared encoder–decoder trunk followed by separate prediction heads for destination and success logits. Both heads operate at full spatial resolution, enabling dense per-cell predictions. The architecture is intentionally kept fixed across experiments to isolate the effects of state representation under partial observability. Differences in model behavior can therefore be attributed directly to changes in the input features rather than architectural variation.

## 5. RESULTS AND CONCLUSION

We evaluate models using a combination of quantitative metrics and qualitative inspection of predicted probability surfaces. Because the goal of this work is to study spatial representations under partial observability, quantitative metrics are used primarily for controlled comparison and sanity checking, while qualitative analysis plays a central role in assessing model behavior and interpretability.

**5.1. Evaluation Metrics.** The model decomposes pass outcomes into destination selection and pass completion conditioned on a destination. For completion, we evaluate the predicted probability at the ground-truth destination cell using negative log-likelihood (Succ@Dest NLL) and area under the ROC curve (Succ@Dest AUC). For destination prediction, we report the negative log-likelihood of the true destination under the predicted spatial distribution (Dest NLL) and Top- $K$  accuracy (Dest Top-10). Because destination selection is inherently multi-modal and discretized, these metrics are interpreted cautiously.

**5.2. Quantitative Ablation Results.** Table 1 summarizes the ablation study across feature groups.

| Ablation setting            | Succ@Dest AUC $\uparrow$ | Succ@Dest NLL $\downarrow$ | Dest Top-10 $\uparrow$ | Dest NLL $\downarrow$ |
|-----------------------------|--------------------------|----------------------------|------------------------|-----------------------|
| Drop vels + gauss           | 0.9847                   | 0.1582                     | 0.0887                 | 6.9308                |
| Drop vels + nearest         | 0.9850                   | 0.1574                     | 0.0786                 | 7.0217                |
| Drop vels + gauss + nearest | 0.9853                   | 0.1540                     | 0.1069                 | 6.8584                |
| Drop static channels        | 0.9972                   | 0.0653                     | 0.2082                 | 5.7871                |
| Drop gauss & nearest        | 0.9967                   | 0.0681                     | 0.2308                 | 5.7305                |
| Drop nearest defender map   | 0.9969                   | 0.0664                     | 0.2336                 | 5.7014                |
| Drop gaussian control map   | <b>0.9972</b>            | <b>0.0642</b>              | <b>0.2425</b>          | <b>5.6965</b>         |
| Drop ball + team velocities | 0.9847                   | 0.1564                     | 0.1288                 | 6.7528                |
| Drop ball velocity only     | 0.9975                   | 0.0602                     | 0.1515                 | 6.5123                |

TABLE 1. Ablation study on the validation set ( $N = 7873$ ). Metrics quantify (i) completion prediction at the ground-truth destination cell (Succ@Dest) and (ii) destination distribution quality (Dest). Quantitative metrics are used primarily as a sanity check and for controlled comparison; qualitative probability surfaces remain the primary evaluation signal in this work.

Across most configurations, completion prediction at the ground-truth destination is uniformly strong, with Succ@Dest AUC values near saturation. This is expected: conditioning on a fixed destination collapses much of the uncertainty in the task, reducing completion prediction to a local geometric problem governed primarily by defender proximity. As a result, even relatively minimal representations achieve high AUC, and differences between feature sets are small. Noticeable degradation occurs only when both motion summaries and interaction-based features are removed, indicating that coarse contextual cues help stabilize feasibility estimates in ambiguous freeze-frame settings.

In contrast, destination prediction metrics exhibit higher variance and are substantially harder to interpret. In several cases, destination Top- $K$  accuracy and negative log-likelihood improve when interaction features or static channels are removed. These apparent gains do not indicate better modeling of pass intent, but instead expose limitations of the supervision and evaluation protocol. Destination prediction is trained under weak spatial supervision and normalized via a dense softmax over the full pitch, making the loss highly sensitive to distributional sharpness and multi-modality. Removing features can therefore encourage broader or less structured distributions that achieve lower loss values without representing more realistic passing behavior. Empirically, the destination loss was harder to optimize than the completion loss and often dominated training dynamics despite being down-weighted. Under these conditions, scalar destination metrics provide only weak quantitative grounding for comparing representations and may reward pathological behaviors such as excessive smoothing.

**5.3. Qualitative Analysis.** We qualitatively compare predictions across model variants on identical pass events to assess how different spatial feature sets influence learned probability surfaces. For each model, we visualize the completion probability surface  $P(\text{completion} \mid s, g)$ , the destination distribution  $P(\text{dest} \mid s)$ , and their product, which we interpret as a relative ranking over candidate pass options. In all figures, the ball is shown in black, attacking players in blue, defenders in orange, and the realized destination in red.

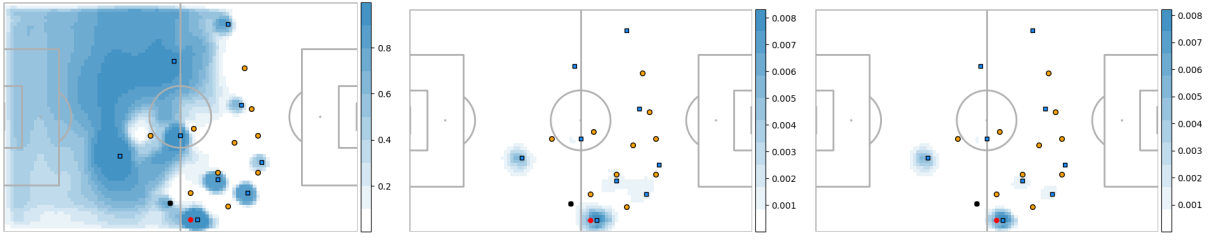


FIGURE 2. Baseline model predictions for a fixed pass event.

Across a wide range of events, the baseline representation, composed only of ball-relative features and binary player occupancy already produces spatially coherent predictions (Figure 2). Completion probabilities concentrate near viable receivers, implausible regions of the pitch are suppressed, and destination probability mass is assigned to a small set of reasonable options. This indicates that a large fraction of pass feasibility and intent can be captured using minimal spatial cues, even under severe partial observability.

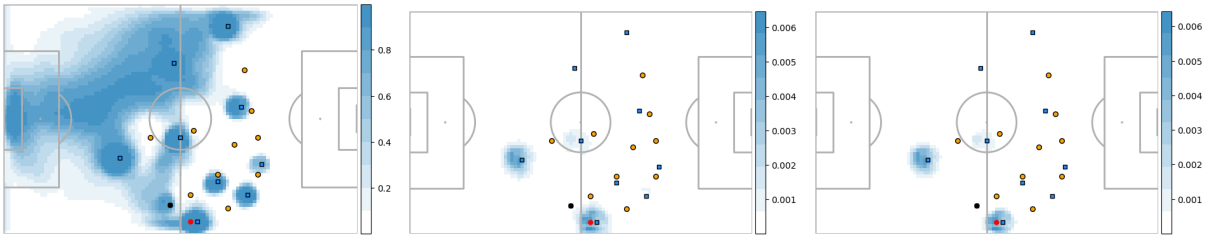


FIGURE 3. Predictions from the final model configuration.

Augmenting the baseline with interaction-based features yields more nuanced behavior in complex situations. For routine passes, predictions from the enriched model closely resemble those of the baseline, suggesting diminishing returns from additional features in simple geometric settings. Differences become most apparent in ambiguous or unsuccessful pass events (Figure 3), where the enriched model more consistently favors alternative destinations with

greater spatial separation from defenders or stronger local control. In these cases, the added features improve the model’s ability to reason about spatial risk and availability, particularly in counterfactual scenarios.

At the same time, the qualitative results expose a limitation of the current formulation. The destination distribution  $P(\text{dest} \mid s)$  is normalized over the full spatial grid via a softmax, resulting in extremely low absolute probability mass at individual locations. With thousands of grid cells, even highly plausible destinations receive very small probabilities, causing the surfaces to appear diffuse or underconfident. While the relative spatial structure remains meaningful, the absolute scale of the probabilities is difficult to interpret and can obscure distinctions between strong and weak candidate actions, especially in complex situations with many plausible options.

**5.4. Conclusion.** This work demonstrates that meaningful and interpretable pass probability surfaces can be learned from sparse, broadcast-derived soccer data using carefully designed spatial representations. Simple baseline features capture much of the structure required for pass feasibility, while a small number of interaction-based features provide additional benefit in more complex or ambiguous situations. At the same time, the results highlight a key limitation of modeling pass destination selection as a dense spatial softmax under partial observability. Addressing this issue, through alternative normalization schemes, region-based predictions, or different formulations of pass intent, represents an important direction for future work. More broadly, these findings underscore the central role of state representation in soccer analytics under realistic observational constraints.

## APPENDIX A. VISUALIZATION OF SPATIAL FEATURE CHANNELS

This appendix provides visualizations of the spatial feature channels and model outputs discussed in the main text. These figures are included for qualitative reference and interpretability but are omitted from the main body for space. All visualizations correspond to examples drawn from the validation set unless otherwise stated.

**A.1. Static Spatial Feature Channels.** Figure 4 visualizes the static spatial channels used in the state representation, including distance-to-boundary maps, distance to the centerline, and goal-oriented encodings such as distance and angle to the attacking goal. These channels encode pitch geometry that is invariant across events and provide a global spatial reference for the model.

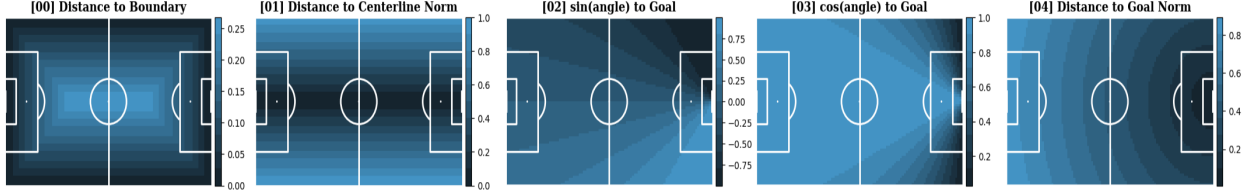


FIGURE 4. Static spatial feature channels used in the state representation.

**A.2. Dynamic Motion Channels.** Figure 5 shows the heuristic motion channels derived from event-to-event displacements, including approximate global attacking and defending team velocity and ball displacement over time. These channels provide weak contextual cues about local motion in the absence of continuous tracking.

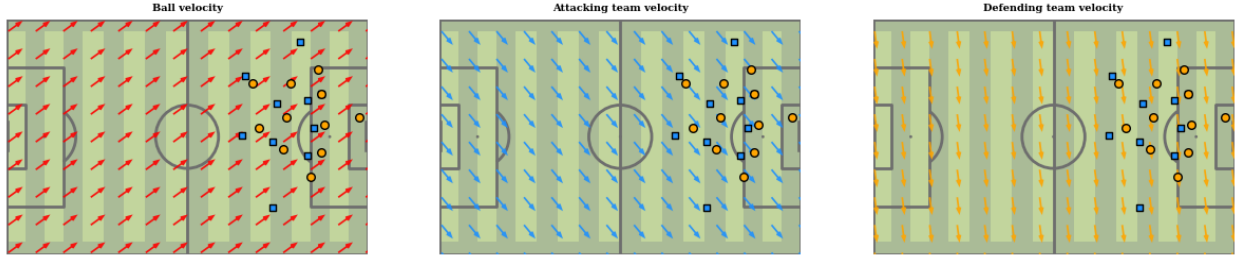


FIGURE 5. Heuristic motion channels derived from event-to-event displacements.

**A.3. Control Maps for Spatial Interaction.** Figure 6 illustrates the two control maps used to encode local spatial interaction under partial observability. The Gaussian control map represents smooth spatial influence via summed player-centered kernels, while the nearest defender distance map encodes local defensive pressure as distance to the closest observed defender.

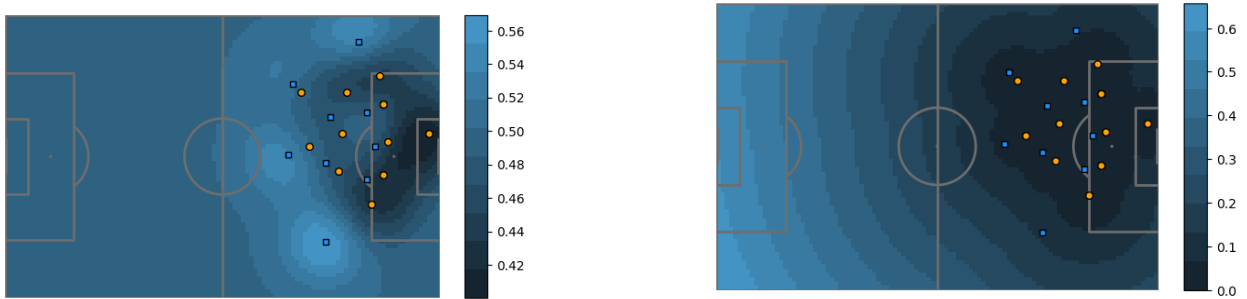


FIGURE 6. Control maps derived from attacker and defender occupancies. Left: Gaussian control map. Right: nearest defender distance map.

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